STACKS, DEFORMATION THEORY: LEMMA-LIFT-COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5267-5311 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 5313-5405, the complete author proof (91 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 7 of queue_defos.json (pb=3417, stmt 5267-5311, proof 5313-5405).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-lift-complex' also exists in stacks-more-algebra — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings. Assume given the following data*

(1) *flat $\mathcal{O}$-modules $\mathcal{G}^n$,*
(2) *maps of $\mathcal{O}$-modules $\mathcal{G}^n \to \mathcal{G}^{n+1}$,*
(3) *a complex $\mathcal{K}_0^\bullet$ of $\mathcal{O}_0$-modules,*
(4) *maps of $\mathcal{O}$-modules $\mathcal{G}^n \to \mathcal{K}_0^n$*

such that

(a) *$H^n(\mathcal{K}_0^\bullet) = 0$ for $n \gg 0$,*
(b) *$\mathcal{G}^n = 0$ for $n \gg 0$,*
(c) *with $\mathcal{G}_0^n = \mathcal{G}^n \otimes_{\mathcal{O}} \mathcal{O}_0$ the induced maps determine a complex $\mathcal{G}_0^\bullet$ and a map of complexes $\mathcal{G}_0^\bullet \to \mathcal{K}_0^\bullet$.*

Then there exist

(i) *flat $\mathcal{O}$-modules $\mathcal{F}^n$,*
(ii) *maps of $\mathcal{O}$-modules $\mathcal{F}^n \to \mathcal{F}^{n+1}$,*
(iii) *maps of $\mathcal{O}$-modules $\mathcal{F}^n \to \mathcal{K}_0^n$,*
(iv) *maps of $\mathcal{O}$-modules $\mathcal{G}^n \to \mathcal{F}^n$,*

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such that $\mathcal{F}^n = 0$ for $n \gg 0$, such that the diagrams

$$\begin{array}{ccc} \mathcal{G}^n & \longrightarrow & \mathcal{G}^{n+1} \\ \downarrow & & \downarrow \\ \mathcal{F}^n & \longrightarrow & \mathcal{F}^{n+1} \end{array}$$

commute for all n, such that the composition $\mathcal{G}^n \to \mathcal{F}^n \to \mathcal{K}_0^n$ is the given map $\mathcal{G}^n \to \mathcal{K}_0^n$, and such that with $\mathcal{F}_0^n = \mathcal{F}^n \otimes_{\mathcal{O}} \mathcal{O}_0$ we obtain a complex $\mathcal{F}_0^\bullet$ and map of complexes $\mathcal{F}_0^\bullet \to \mathcal{K}_0^\bullet$ which is a quasi-isomorphism.

Proof. We will prove by descending induction on e that we can find $\mathcal{F}^n$, $\mathcal{G}^n \to \mathcal{F}^n$, and $\mathcal{F}^n \to \mathcal{F}^{n+1}$ for $n \geq e$ fitting into a commutative diagram

$$\begin{array}{ccccccccc} \ldots & \longrightarrow & \mathcal{G}^{e-1} & \longrightarrow & \mathcal{G}^e & \longrightarrow & \mathcal{G}^{e+1} & \longrightarrow & \ldots \\ & & & & \downarrow & & \downarrow & & \\ & & & & \mathcal{F}^e & \longrightarrow & \mathcal{F}^{e+1} & \longrightarrow & \ldots \\ & & & & \downarrow & & \downarrow & & \\ \ldots & \longrightarrow & \mathcal{K}_0^{e-1} & \longrightarrow & \mathcal{K}_0^e & \longrightarrow & \mathcal{K}_0^{e+1} & \longrightarrow & \ldots \end{array}$$

such that $\mathcal{F}_0^\bullet$ is a complex, the induced map $\mathcal{F}_0^\bullet \to \mathcal{K}_0^\bullet$ induces an isomorphism on H^n for $n > e$ and a surjection for $n = e$. For $e \gg 0$ this is true because we can take $\mathcal{F}^n = 0$ for $n \geq e$ in that case by assumptions (a) and (b).

Induction step. We have to construct $\mathcal{F}^{e-1}$ and the maps $\mathcal{G}^{e-1} \to \mathcal{F}^{e-1}$, $\mathcal{F}^{e-1} \to \mathcal{F}^e$, and $\mathcal{F}^{e-1} \to \mathcal{K}_0^{e-1}$. We will choose $\mathcal{F}^{e-1} = A \oplus B \oplus C$ as a direct sum of three pieces.

For the first we take $A = \mathcal{G}^{e-1}$ and we choose our map $\mathcal{G}^{e-1} \to \mathcal{F}^{e-1}$ to be the inclusion of the first summand. The maps $A \to \mathcal{K}_0^{e-1}$ and $A \to \mathcal{F}^e$ will be the obvious ones.

To choose B we consider the surjection (by induction hypothesis)

$$\gamma : \mathrm{Ker}(\mathcal{F}_0^e \to \mathcal{F}_0^{e+1}) \longrightarrow \mathrm{Ker}(\mathcal{K}_0^e \to \mathcal{K}_0^{e+1}) / \mathrm{Im}(\mathcal{K}_0^{e-1} \to \mathcal{K}_0^e)$$

We can choose a set I, for each $i \in I$ an object U_i of $\mathcal{C}$, and sections $s_i \in \mathcal{F}^e(U_i)$, $t_i \in \mathcal{K}_0^{e-1}(U_i)$ such that

(1) s_i maps to a section of $\mathrm{Ker}(\gamma) \subset \mathrm{Ker}(\mathcal{F}_0^e \to \mathcal{F}_0^{e+1})$,
(2) s_i and t_i map to the same section of $\mathcal{K}_0^e$,
(3) the sections s_i generate $\mathrm{Ker}(\gamma)$ as an $\mathcal{O}_0$-module.

We omit giving the full justification for this; one uses that $\mathcal{F}^e \to \mathcal{F}_0^e$ is a surjective maps of sheaves of sets. Then we set to put

$$B = \bigoplus\nolimits_{i \in I} j_{U_i!}\mathcal{O}_{U_i}$$

and define the maps $B \to \mathcal{F}^e$ and $B \to \mathcal{K}_0^{e-1}$ by using s_i and t_i to determine where to send the summand $j_{U_i!}\mathcal{O}_{U_i}$.

With $\mathcal{F}^{e-1} = A \oplus B$ and maps as above, this produces a diagram as above for $e-1$ such that $\mathcal{F}_0^\bullet \to \mathcal{K}_0^\bullet$ induces an isomorphism on H^n for $n \geq e$. To get the map to be surjective on H^{e-1} we choose the summand C as follows. Choose a set J, for each $j \in J$ an object U_j of $\mathcal{C}$ and a section t_j of $\mathrm{Ker}(\mathcal{K}_0^{e-1} \to \mathcal{K}_0^e)$ over U_j such that these sections generate this kernel over $\mathcal{O}_0$. Then we put

$$C = \bigoplus\nolimits_{j \in J} j_{U_j!}\mathcal{O}_{U_j}$$

and the zero map $C \to \mathcal{F}^e$ and the map $C \to \mathcal{K}_0^{e-1}$ by using s_j to determine where to the summand $j_{U_j!}\mathcal{O}_{U_j}$. This finishes the induction step by taking $\mathcal{F}^{e-1} = A \oplus B \oplus C$ and maps as indicated. $\square$